

Red Tent Academy

01) KPI Dictionary & Data Quality Contract.

→ Translate an ambiguous business request into measurable definitions & a testable agreement about what trustworthy data mean.

→ Before running complex analytics, a data team must establish two core guidelines.

a) The KPI Dictionary → What exact business equations & numbers are we ~~measuring~~ measuring?

b) Data Quality Contract → What are the rules that incoming raw data must satisfy & what happens when bad data breaches those rules?

⊕ Core Concept to Learn / Master.

A) A Decision Owner:

→ What it mean - The specific business leader or executive who depend on the ~~business~~ matrix to make strategic decisions.

eg:- Head of E-commerce owns - Revenue matrix.

Customer Operation Lead owns - Repeat / Fillment matrix.

B) The 5 Data Quality Dimensions.

→ While inspecting any dataset, test for these 5 dimensions.

1) Completeness → Are any mandatory fields missing / null?

2) Uniqueness → Are there unexpected duplicate records?

3) Validity → Do values conform to allowable formats & range?

4) Consistency → Are text casing, dates & data types uniform?

5) Freshness → Is the data up to date & within expected historical bound?

7 Data Quality Control Structure:

- Rules - "field 'quantity' must be integer > 0."
- Threshold - "0% bad rows allowed (Zero tolerance)"
- Escalation Action - "Flag & quarantine bad rows; send an alert to the Data Engineering team."

KPI ???

→ Key Performance Indicator.

→ KPI are just important numbers you track regularly to measure success.

- eg:
- Scoreboard show goal scored - In football (KPI per match)
 - Exam % (KPI of your study)
 - No. like Revenue, Profit, Customer Satisfaction or Delivery on time (KPI in business)

→ While shifting into data analytics, working with tons of data - KPI help focus on the numbers that actually matter. Instead of drowning in all the details.

- eg:
- Gross Revenue - Total money earned.
 - Net Revenue - Money left after costs.
 - Avg. Order Value - How much customer spent per order.



Pandas & NumPy

→ Two core Python library for data analysis.

7 NumPy (Numerical Python)

- High performance numerical computing
- ndarray (N-dimensional array) strong elements of the same type in contiguous memory.
- Best for numerical data, scientific computing & performing critical tasks.

67 Pandas:

- Data manipulation & analysis with tabular data.
- Data Frame (2D table) & Series (1D labeled array)
- {
 - ↳ Import/Export from CSV, Excel, SQL, JSON
 - ↳ Data cleaning (handling missing values, duplicates, inconsistent formats)
 - ↳ Grouping & aggregation (groupby)
 - ↳ Time Series handling}
- Best for - Real world datasets (like retail orders, financial records, sensor logs) where labels columns & indices matter

`import pandas as pd`
`import numpy as np`

- Similar in spirit to `#include <stdio.h>` in C, but not exactly same
- `import pandas` / `import numpy` - Loads the external library so I can use its functions.
- `as pd` / `as np` - Giving each library a unique and easy to use nickname (alias)
- ⊙ It's like including a header in C, Python imports are much powerful - they load entire modules with functions, classes & objects.
- ⊙ `np.array` → comes from numpy.
`pd.DataFrame` → comes from pandas.
 - { You must prefix the function with the library alias so Python knows where it belongs.

df = pd.read_csv ("file location / filename.csv")

pd.read_csv () → Pandas function that reads csv files & loads it into memory as a DataFrame.

⇒ A dataframe in pandas is basically a 2D labeled table data - very much like Excel sheet or a SQL table, but living in Python's memory.

⇒ "<code>" → The <code>-makes it raw string, <code>() are treated literally (Important for window path)